

SHI: Inspecting and Correcting Structured Hypotheses in Interactive Video Retrieval

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Abstract. Large language models can decompose complex video-search narratives into ordered events, but errors in event boundaries or temporal relations may affect downstream retrieval. In multi-event search, such errors are difficult to diagnose when both query interpretation and temporal alignment remain internal to the retrieval pipeline. We present **SHI**, an interactive video retrieval system with two interaction stages: *pre-retrieval* and *post-retrieval*. In the *pre-retrieval* stage, the *Query Hypothesis* exposes the LLM-generated decomposition as an editable ordered sequence of events, allowing users to inspect and revise the search structure before retrieval. In the *post-retrieval* stage, *EventTrail* exposes the decoded temporal alignment of a candidate video and presents alternative event occurrences as complete ordered paths rather than isolated frames. Users can anchor accepted occurrences or exclude incorrect temporal regions, after which the alignment is re-decoded under the resulting constraints. Because re-decoding operates on retained retrieval evidence, these corrections can be applied without repeating corpus-level multimodal retrieval. Together, the two stages make both semantic decomposition and temporal alignment explicit interaction objects in multi-event video search.

Keywords: Interactive video retrieval · Video Browser Showdown · Temporal alignment · Human-in-the-loop · Multi-event search

1 Introduction

The Video Browser Showdown (VBS) evaluates interactive video retrieval systems through live search tasks over large-scale video collections [16]. Among these tasks, multi-event Known-Item Search (KIS) requires an operator to locate a video described by a sequence of temporally ordered events [16]. Compared with single-event retrieval, this setting introduces two additional sources of uncertainty: the narrative must first be decomposed into suitable retrieval events, and the retrieved evidence must then be aligned to occurrences that satisfy the intended temporal order. Errors in either stage can affect candidate ranking and complicate subsequent inspection. However, these intermediate decisions are often handled internally by the retrieval pipeline, leaving operators to infer how a narrative was interpreted or why particular temporal occurrences were selected from the returned results [1, 8, 14, 3, 6].

LLMs provide a convenient way to convert unstructured narratives into structured event sequences. Prior work on complex task understanding shows that LLMs can identify individual task steps effectively, while still exhibiting difficulty in predicting temporal dependencies between those steps [18]. This distinction is particularly relevant to multi-event video search, where an apparently plausible decomposition may still contain event boundaries or ordering assumptions that do not match the operator’s intended search strategy. If such a decomposition is executed directly, its errors may propagate to multimodal scoring and temporal alignment. We therefore treat the LLM output as a search hypothesis rather than a fixed interpretation and expose it for inspection and editing before retrieval.

Interaction has repeatedly been identified as an important component of VBS search performance. A post-evaluation of VBS 2022 found that strong CLIP-based text retrieval benefits KIS, but needs to be complemented by effective browsing and additional query modalities for consistent performance [13]. More recent interaction logs from VBS 2026 show a strong reliance on iterative, high-frequency text-query reformulation coupled with result-set inspection [4]. Existing interactive video retrieval systems provide rich mechanisms for query formulation, browsing, and relevance feedback [1, 8]; however, for multi-event search, the decoded temporal alignment of a candidate is typically not exposed as an object that the operator can directly inspect and modify. Consequently, when several events are matched correctly but one occurrence is temporally misaligned, the operator may need to reformulate the query or continue browsing despite already having useful retrieval evidence. This motivates interaction not only with the input query, but also with the temporal hypothesis produced for an individual candidate.

To address these two interaction points, we present **SHI**, an interactive multimodal video retrieval system for multi-event KIS with complementary *pre-retrieval* and *post-retrieval* mechanisms. **Pre-retrieval Query Hypothesis** exposes the LLM-generated event decomposition as an editable structured plan, allowing operators to inspect event boundaries, revise event descriptions, reorder events, and attach visual reference examples before search. **Post-retrieval EventTrail** exposes the decoded temporal alignment of a candidate video as an ordered event path, allowing operators to inspect alternative occurrences, anchor accepted matches, and exclude incorrect temporal regions. The resulting constraints are applied by re-decoding the alignment from retained retrieval evidence without repeating corpus-level retrieval. Together, these mechanisms make semantic decomposition before retrieval and temporal alignment after retrieval explicit parts of the interactive search process.

2 Related Work

Interactive video retrieval systems developed for the Video Browser Showdown (VBS) combine multimodal retrieval with mechanisms for query formulation, iterative refinement, and result exploration. VISIONE supports textual, visual-example, and sketch-based queries [1], while PraK provides interactive search and filtering over video data services [8]. Exquisitor further combines conversational search with user relevance feedback in a unified retrieval interface [14]. Evaluations of VBS systems have repeatedly highlighted the importance of iterative query reformulation and result inspection in

addition to the quality of the initial ranking [7, 13, 4]. These systems provide rich mechanisms for modifying queries and exploring retrieved results, but the decomposition of a narrative request into an ordered sequence of retrieval events is generally not exposed as an editable intermediate representation [1, 8, 14]. SHI addresses this interaction point through the *Query Hypothesis*, which presents the event decomposition explicitly and allows operators to modify its structure before retrieval.

Temporal structure has also been incorporated into interactive video retrieval in several forms. Vireo-VSS extends its SQL-like interface with templates for specifying temporal query evolution [17], while vitrivr supports multimodal queries with temporal context [3]. More recently, Exquisitor introduced sequence-chain retrieval for temporal queries [6]. Interactive video corpus moment retrieval has also used user feedback to refine the retrieval of temporal intervals [9]. These approaches incorporate temporal structure through query specification, retrieval, or feedback over candidate intervals. SHI focuses on a complementary interaction point: the decoded multi-event alignment of an individual candidate video. Its *EventTrail* exposes this alignment as an ordered path, allows operators to anchor or reject individual event occurrences, and re-decodes the path under the resulting temporal constraints using retained retrieval evidence.

3 System Overview

SHI separates offline corpus preparation from the online retrieval workflow. During offline preparation, each target video collection is processed into canonical frame meta-data and precomputed retrieval artifacts, including dense FAISS indexes [5] for visual, contextual, and speech evidence and a lexical BM25 index. At runtime, as illustrated in Fig. 1, these artifacts are loaded as read-only retrieval structures and queried for each event in the active Query Hypothesis. Corpus-side feature extraction and index construction therefore remain outside the interactive request path, while online processing is limited to query encoding, retrieval, score fusion, and temporal alignment.

3.1 Multi-event KIS Retrieval Pipeline

Multi-event KIS is formulated as multimodal event scoring followed by joint temporal alignment. Given an ordered sequence of events from the Query Hypothesis, each textual event is evaluated against several retrieval sources: SigLIP2 [15] provides visual text–frame similarity, BGE-M3 [2] provides dense FrameContext and ASR evidence, and BM25 [11] provides lexical matching over frame-associated text. Optional reference images contribute additional SigLIP2 image–frame similarity scores. The resulting components are calibrated and combined by event-adaptive weighted fusion into an event–frame score matrix $S[v, e, f]$ for each candidate video v . A dynamic-programming decoder then selects one canonical frame occurrence for each event subject to strict temporal ordering. Its objective combines the fused event–frame scores with a temporal gap penalty that discourages large separations between consecutive event occurrences. Candidate videos are ranked by the scores of their decoded paths, so the ranking accounts jointly for event-level evidence and temporal ordering rather than aggregating independently selected frame matches.

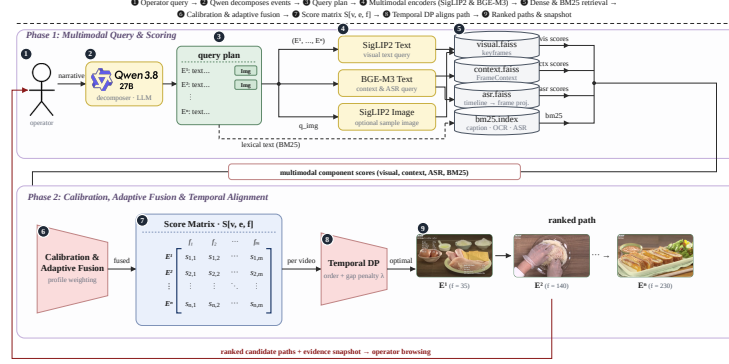


Fig. 1. Multi-event KIS retrieval pipeline in SHI. The Query Hypothesis provides an ordered sequence of events. Text events query visual, FrameContext, ASR, and lexical indexes, while optional reference images contribute visual similarity scores. Component scores are calibrated and fused into an event–frame score matrix $S[v, e, f]$ for each candidate video v . A dynamic-programming decoder then extracts chronologically ordered event paths, which are used to rank candidate videos and construct the evidence snapshot.

3.2 Evidence Snapshots and Competition Integration

To support post-retrieval interaction without repeating corpus-level search, SHI captures an in-memory *Evidence Snapshot* after the initial KIS execution. For the returned candidates, the snapshot retains immutable event–frame evidence together with the event identifiers, initial decoded paths, and the decoder configuration used for that search. When an operator applies an EventTrail action, such as anchoring an accepted occurrence or rejecting a temporal region, the corresponding constraints are maintained separately from the frozen evidence and supplied to the temporal decoder. The decoder then re-computes the feasible path for the selected video using the retained evidence, without repeating query embedding or corpus index search. Once the operator selects a frame for submission, SHI’s integrated competition client converts the selection to the required submission format and sends it to the Distributed Retrieval Evaluation Server (DRES) [12], reducing the need for manual transfer of frame coordinates during live search.

4 Interactive Hypothesis Inspection and Trajectory Correction

SHI exposes two intermediate representations for direct interaction during multi-event search. In the *pre-retrieval* stage, the Query Hypothesis represents the current interpretation of a narrative as an ordered sequence of events that can be inspected and edited before search. In the *post-retrieval* stage, EventTrail represents the temporal alignment of an individual candidate video as an ordered path over event occurrences. As illustrated in Fig. 2, edits to the Query Hypothesis determine the event specifications used by retrieval, whereas EventTrail actions are translated into temporal constraints and

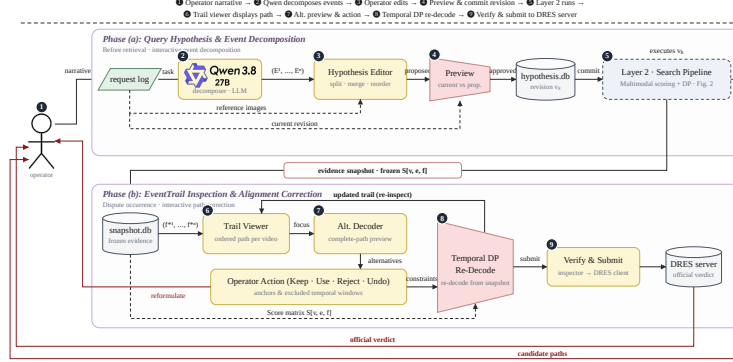


Fig. 2. The two hypothesis-level interaction mechanisms in SHI. (a) Query Hypothesis exposes the initial LLM-derived event decomposition before retrieval and supports structural editing, reference-image binding, and non-destructive preview before a revision is committed. (b) Event-Trail exposes the decoded temporal path of a candidate video and presents temporally distinct alternatives as complete ordered paths. Keep, Use, and Reject actions introduce anchor or exclusion constraints, after which the path is re-decoded from the retained evidence snapshot.

re-decoded over retained retrieval evidence. The two stages therefore expose semantic decomposition and temporal alignment as separate but connected interaction points in the retrieval workflow.

4.1 Pre-retrieval Query Hypothesis

The Query Hypothesis represents the current semantic interpretation of a multi-event request before retrieval is executed. Given a narrative query, the LLM (Qwen [10]) proposes an ordered sequence of events $(E_1, \dots, E_N)$. For the initial decomposition, each proposed event must correspond to a verbatim fragment of the original query; the server verifies these fragments against the source text and falls back to a single-event representation if the decomposition cannot be grounded. The resulting event sequence is shown explicitly to the operator together with its temporal order and any event-specific reference images. Rather than treating the decomposition as a fixed parser output, SHI keeps it as the active search hypothesis from which the downstream retrieval plan is constructed.

The editor supports event-level structural operations including splitting and merging events, changing their order, editing or adding event descriptions, and attaching or removing reference images. Each proposed operation can first be previewed without modifying the active hypothesis, allowing the operator to compare the current and proposed event sequences before committing the change. A committed operation creates a new monotonically revisioned hypothesis and stores the previous state in a bounded session history, from which the latest change can be undone. Subsequent retrieval uses the committed revision, so structural edits directly change the event specifications supplied to the multimodal scoring and temporal-alignment pipeline.

4.2 Post-retrieval EventTrail

EventTrail operates on the decoded temporal path of an individual candidate video after the initial retrieval. For a query with N events, the trail contains one selected frame occurrence for each event in temporal order, together with the corresponding retained event–frame evidence. An operator can focus on any event whose current occurrence appears inconsistent with the video content. EventTrail then identifies temporally distinct alternative occurrences for that event and decodes a complete path for each alternative. The interface therefore presents the effect of an alternative on the entire event sequence rather than showing the alternative frame in isolation. These previews do not modify the active trail until the operator explicitly commits an action.

Three actions are provided for occurrence-level correction. *Keep* anchors the occurrence currently assigned to an event, *Use* anchors the representative occurrence of a selected alternative path, and *Reject* excludes the temporal interval associated with a selected occurrence mode. These actions update the session’s temporal constraints while leaving the underlying Evidence Snapshot unchanged. The dynamic-programming decoder then re-computes the highest-scoring feasible path for the selected video under the updated anchors and exclusion intervals. Because the event–frame evidence is retained in the snapshot, this correction step does not repeat query embedding or corpus-level retrieval. EventTrail also maintains a bounded history of constraint states, allowing the most recent action to be undone before further inspection or submission.

5 Conclusion

We presented **SHI**, an interactive multimodal video retrieval system for multi-event Known-Item Search at VBS 2027. SHI exposes two intermediate retrieval hypotheses for direct user interaction. The pre-retrieval *Query Hypothesis* represents the LLM-generated event decomposition as an editable ordered sequence, allowing operators to inspect and revise the search structure before retrieval. The post-retrieval *Event-Trail* exposes the decoded temporal alignment of an individual candidate video and supports occurrence-level correction through anchor and exclusion constraints. By retaining event–frame evidence in session-scoped Evidence Snapshots, SHI can re-decode a constrained candidate path without repeating corpus-level retrieval. Together, these mechanisms make semantic decomposition before retrieval and temporal alignment after retrieval explicit interaction points in the multi-event search workflow.

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